



Rapid Response Mold Testing

Prepared By:
Rapid Response Mold Testing

+92 03168614915

Moldtesting@gmail.com

www.company.com

MOLD INSPECTION REPORT



Prepared For:

Shahbaz

Property Address:

24.880486, 67.032665

Date of Inspection:

July 23, 2026

Inspector:

Inspector



TABLE OF CONTENTS

I. Building Profile	3
II. Scope of Inspection	3
III. Inspection Description	4
IV. Observations and Readings	4
V. Methodology	6
VI. Scope of Work Summary	7
VII. Laboratory Results	7
VIII. Conclusion	8
IX. Recommendations	9
X. Laboratory Information	9
XI. Limitations	10
XII. Laboratory Report (Attachment)	12



I. BUILDING PROFILE

The subject property is located at **24.880486, 67.032665**. The structure is classified as a Residential. The building was constructed in 2004. The approximate square footage is 1200 sq ft.

II. SCOPE OF INSPECTION

This inspection was conducted to assess the property for visible mold growth, moisture intrusion, and conditions conducive to mold development. The inspection included visual examination of accessible areas, moisture meter readings, and sample collection for laboratory analysis where conditions warranted.

The inspection was performed on **July 23, 2026** by **Inspector**. All findings are based on conditions observed at the time of inspection.

III. VISUAL INSPECTION

The following areas were inspected for moisture intrusion, water damage, and potential mold growth:

Room1

Moisture Reading:	25%
Material Tested:	Drywall
Water Damage:	Staining, Bubbling
Odor Presence:	Musty
Thermal Imaging:	Cold spots detected
Inspector Notes:	This is the room one description

Thermal Imaging Documentation:



1



2



3



Moisture Meter Photo Documentation:



1



2



3

IV. OBSERVATIONS AND READINGS

The following areas were inspected and documented for visible conditions, moisture levels, and environmental factors:

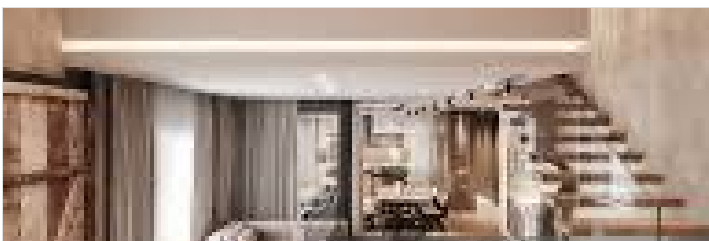
Room1

Observed Conditions:

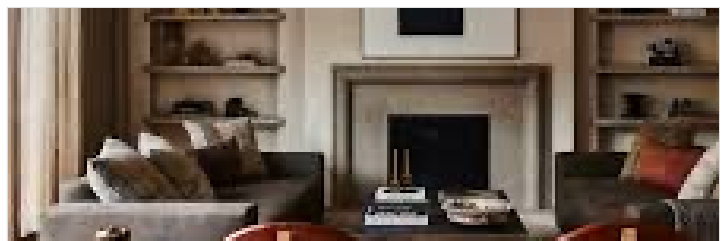
- Growth type: Spotting, Fuzzy
- Color/appearance: Black
- Surface affected: Carpet
- Condition: Old / Staining
- Approximate size: 1–10 sq ft

Inspector Notes: This is the room one description

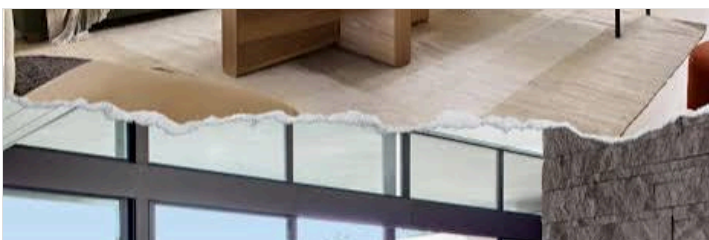
Photo Documentation:



Room1



Roo2



Room3



ON-SITE MOISTURE READINGS

Area	Material	Reading	Status
Room1	Drywall	25%	Elevated

Moisture readings were obtained using a moisture meter on both surface (non-invasive) and penetrating modes. The readings documented in the on-site table were compared against industry standard moisture thresholds to evaluate the potential for mold growth.

AIR SAMPLE SUMMARY

In a properly maintained structure, indoor fungal concentrations are typically similar to or lower than outdoor levels. Indoor air samples should be compared against outdoor baseline samples to identify elevated conditions.

Spore Trap Results (spores/m³)

Location	Total Spores/m ³	Pen/Asp	Cladosporium	Ascospores	Basidiospores	Alternaria	Chaetomium	Stachybotrys	Status
Master Bedroom	2840	267	1973	8	267	213	—	—	Elevated
Backyard	240	53	12	0	160	53	—	—	

Understanding "Elevated" Status:

The status determination is based on **total spore count comparison** between the indoor containment area and the outdoor baseline sample. This methodology compares the overall fungal burden in both environments to determine if remediation has successfully returned the indoor space to normal background conditions.

Status Criteria:

- **"Elevated"** – Assigned when the indoor total spore count **exceeds** the outdoor baseline count, indicating amplification of fungal spores within the indoor environment.
- **"Normal"** – Assigned when the indoor total spore count is **equal to or lower than** the outdoor baseline, demonstrating that indoor conditions are consistent with normal environmental levels.

Example from this report: Indoor Total = **spores/m³**, Outdoor Baseline = **spores/m³**. Since the indoor count () is lower than the outdoor baseline (), the status is assigned as **"Normal"**, indicating the containment area has returned to acceptable background conditions.

Note: While individual fungal species may show different patterns between indoor and outdoor samples, the overall status determination is based on the total airborne spore concentration comparison, which provides a comprehensive assessment of the indoor air quality relative to normal outdoor conditions.

Detailed Sample Information

Sample SMP-1784720025201 — Living room

Sample ID:	SMP-1784720025201
Location:	Living room
Sample Type:	Air
Collection Method:	Spore Trap (AirCassette)
Flow Rate:	25 L/min
Duration:	30 min
Total Volume:	750.00 L



Chain of Custody:	COC ID: H-260623102
Inspector Notes:	This is living room samples

Sample Location Photos:



Sample Location Photo - Sample1



Sample Location Photo - Sample2



Sample Location Photo - Sample3



Sample Location Photo - Sample4

Mold Close-Up Photo Documentation:



1



2



3



4

Southeast Environmental Microbiology Laboratories

Location Comparison	Total Spores/m ³	Dominant Species	Status
Master Bedroom	2840	Cladosporium	Elevated
Backyard	240	Basidiospores	

V. METHODOLOGY

Moisture Assessment:



- Non-invasive moisture meter readings were obtained at the following 1 location(s): Room1 (Drywall).
- Moisture readings ranged from 25% to 25%.
- Readings were compared against industry standard moisture thresholds (dry: <15%, moderate: 15-20%, elevated: >20%).

Air Sampling:

- Air samples were collected using spore trap cassettes at the following locations:
 - **Living room** - Sample ID: SMP-1784720025201, Flow rate: 25 L/min, Duration: 30 min
- Samples were analyzed via direct microscopic examination for spore identification and quantification.

Laboratory Analysis:

- All samples were submitted to **Southeast Environmental Microbiology Laboratories** for analysis.
- Spore trap analysis included identification and quantification of fungal spores per cubic meter of air (spores/m³).
- Surface sample analysis identified fungal structures, hyphae presence, and general impression of contamination level.
- Indoor air samples were compared against outdoor baseline levels to determine elevation status.

VI. SCOPE OF WORK SUMMARY

Area	Findings	Recommendation
Room1 — Drywall	Elevated moisture detected (25%). Water damage observed: Staining, Bubbling. Musty odor present.	Identify and eliminate moisture source. Implement moisture control measures. Consider mold remediation if visible growth present. Repair water-damaged materials. Ensure proper drainage.
Room1 — Carpet	Growth type: Spotting, Fuzzy (Black). Condition: Old / Staining. Approximate size: 1–10 sq ft. This is the room one description	Professional mold remediation required. Remove contaminated materials following IICRC S520 standards. Address underlying moisture source.

VII. LABORATORY RESULTS

Southeast Environmental Microbiology Laboratories

Lab Reference: H-260623102 Date Reported: 06/23/2026 Date Sampled: 06/20/2026

Air Sample Analysis Results

Sample Location	Total Spores	Pen/Asp	Cladosporium	Ascospores	Basidiospores	Alternaria	Chaetomium	Stachybotrys	Status
Master Bedroom	2840	267	1973	8	267	213	—	—	Elevated
Backyard	240	53	12	0	160	53	—	—	

Note: Indoor air samples are compared against outdoor baseline. Elevated status indicates indoor spore levels significantly exceed outdoor levels, suggesting an indoor amplification source.

Laboratory Findings Interpretation**Master Bedroom (Spore_trap)****Alternaria — 213 spores/m³**

Alternaria is a common outdoor mold that can colonize carpets, drywall, and window frames indoors when moisture is present. It is a highly reactive allergen and a recognized trigger for severe asthma attacks.

Ascospores — 8 spores/m³



Ascospores are primarily outdoor spores associated with rain, dew, and agricultural environments. Their presence indoors is typically due to outdoor air entry and is not an indicator of indoor growth unless standing water is present.

- **Basidiospores — 267 spores/m³**

Basidiospores originate from outdoor mushrooms and bracket fungi. Their presence indoors is generally normal unless associated with structural wood rot or decay.

- **Bipolaris/Drechslera — 13 spores/m³**

Typical environmental mold. Elevated levels indoors relative to outdoor baseline suggest an indoor moisture source. Can contribute to allergic reactions and respiratory irritation.

- **Cladosporium — 1973 spores/m³**

Cladosporium is one of the most widespread environmental molds. Indoors, it commonly grows on damp surfaces, textiles, window frames, and HVAC vents. It is a major allergen and trigger for asthma, dry cough, and skin irritation.

- **Epicoccum — 27 spores/m³**

Typical environmental mold. Elevated levels indoors relative to outdoor baseline suggest an indoor moisture source. Can contribute to allergic reactions and respiratory irritation.

- **Penicillium/Aspergillus — 267 spores/m³**

Penicillium/Aspergillus are common indoor molds that grow rapidly under elevated relative humidity (above 60%) or light moisture. Elevated indoor levels compared to the outdoor baseline indicate active indoor growth and amplification, which can trigger respiratory allergies, asthma, and sinus inflammation.

- **Tetraploa — 27 spores/m³**

Typical environmental mold. Elevated levels indoors relative to outdoor baseline suggest an indoor moisture source. Can contribute to allergic reactions and respiratory irritation.

- **Torula — 40 spores/m³**

Typical environmental mold. Elevated levels indoors relative to outdoor baseline suggest an indoor moisture source. Can contribute to allergic reactions and respiratory irritation.

Backyard (Spore_trap)

- **Alternaria — 53 spores/m³**

Alternaria is a common outdoor mold that can colonize carpets, drywall, and window frames indoors when moisture is present. It is a highly reactive allergen and a recognized trigger for severe asthma attacks.

- **Ascospores — 0 spores/m³**

Ascospores are primarily outdoor spores associated with rain, dew, and agricultural environments. Their presence indoors is typically due to outdoor air entry and is not an indicator of indoor growth unless standing water is present.

- **Basidiospores — 160 spores/m³**

Basidiospores originate from outdoor mushrooms and bracket fungi. Their presence indoors is generally normal unless associated with structural wood rot or decay.

- **Bipolaris/Drechslera — 0 spores/m³**

Typical environmental mold. Elevated levels indoors relative to outdoor baseline suggest an indoor moisture source. Can contribute to allergic reactions and respiratory irritation.

- **Cladosporium — 12 spores/m³**

Cladosporium is one of the most widespread environmental molds. Indoors, it commonly grows on damp surfaces, textiles, window frames, and HVAC vents. It is a major allergen and trigger for asthma, dry cough, and skin irritation.

- **Epicoccum — 27 spores/m³**

Typical environmental mold. Elevated levels indoors relative to outdoor baseline suggest an indoor moisture source. Can contribute to allergic reactions and respiratory irritation.

- **Penicillium/Aspergillus — 53 spores/m³**

Penicillium/Aspergillus are common indoor molds that grow rapidly under elevated relative humidity (above 60%) or light moisture. Elevated indoor levels compared to the outdoor baseline indicate active indoor growth and amplification, which can trigger respiratory allergies, asthma, and sinus inflammation.

- **Tetraploa — 27 spores/m³**

Typical environmental mold. Elevated levels indoors relative to outdoor baseline suggest an indoor moisture source. Can contribute to allergic reactions and respiratory irritation.

- **Torula — 40 spores/m³**

Typical environmental mold. Elevated levels indoors relative to outdoor baseline suggest an indoor moisture source. Can contribute to allergic reactions and respiratory irritation.

Indoor vs. Outdoor Air Comparison & Interpretation:

Fungal spore levels from indoor locations were evaluated against the outdoor baseline sample. A comparison of total spore concentration and individual genus distributions was performed:

- **Master Bedroom (2840 spores/m³):** Spore count is **elevated** relative to the outdoor baseline, indicating indoor amplification or stagnation. Dominant species include Cladosporium.



VIII. CONCLUSION

FAILED CLEARANCE / FAILED MOLD INSPECTION

Based on laboratory findings, the containment area **did not pass clearance testing**.

Analysis of Laboratory Findings:

Air sampling results indicate elevated fungal spore concentrations within the containment area. The outdoor baseline sample measured , while the indoor containment sample measured **2840 spores/m³**, representing an indoor concentration that is elevated compared to the outdoor baseline.

Detailed analysis of individual fungal species reveals:

- **Alternaria:** 213 spores/m³ indoors
- **Ascospores:** 8 spores/m³ indoors
- **Basidiospores:** 267 spores/m³ indoors
- **Bipolaris/Drechslera:** 13 spores/m³ indoors
- **Cladosporium:** 1973 spores/m³ indoors (**Elevated**)
- **Epicoccum:** 27 spores/m³ indoors
- **Penicillium/Aspergillus:** 267 spores/m³ indoors
- **Tetraploa:** 27 spores/m³ indoors
- **Torula:** 40 spores/m³ indoors

Scientific Interpretation:

The elevated indoor spore concentrations, particularly of indicator species such as Penicillium/Aspergillus, are consistent with the presence of an **active indoor fungal reservoir**. This suggests that fungal growth continues within building materials, hidden cavities, or other concealed areas within the containment zone.

When indoor spore levels exceed outdoor baseline concentrations, it indicates that the indoor environment is amplifying fungal spore production beyond normal background levels. This amplification effect demonstrates that the **fungal ecology has not returned to normal conditions** and active colonization remains present.

The persistence of elevated spore levels following remediation activities suggests one or more of the following conditions:

- Incomplete removal of contaminated materials
- Hidden or inaccessible fungal growth not addressed during remediation
- Ongoing moisture intrusion enabling continued fungal amplification
- Inadequate cleaning and HEPA vacuuming of affected surfaces
- Cross-contamination from adjacent unremediated areas

Final Determination:

Based on the laboratory analysis and interpretation of findings, the containment area has **failed post-remediation verification testing**. The indoor environment does not meet acceptable clearance criteria as defined by industry guidelines (IICRC S520, AIHA, ACGIH).

Re-evaluation and additional remediation measures are required before the area can be considered safe for unrestricted occupancy. The property owner should engage qualified remediation professionals to investigate the source of continued fungal amplification and implement appropriate corrective actions.

IX. RECOMMENDATIONS

The following corrective actions are recommended to address the failed clearance and restore the indoor environment to acceptable conditions:



- **Investigate Hidden Fungal Reservoirs:** Further investigate the containment area for hidden fungal growth or remaining contaminated materials that may not have been identified during initial remediation. This includes inspection of wall cavities, ceiling plenums, HVAC ductwork, and any concealed spaces where moisture intrusion or fungal colonization may persist.
- **Re-evaluate Remediated Materials:** Re-evaluate all previously remediated building materials and adjacent wall, ceiling, and flooring assemblies. Materials showing signs of water damage, staining, or elevated moisture content should be considered for removal or additional treatment. Pay particular attention to porous materials (drywall, insulation, ceiling tiles) that may retain fungal contamination even after surface cleaning.
- **Moisture Source Control:** Verify that all moisture sources have been identified and corrected. Conduct moisture mapping using calibrated meters to identify any areas with elevated moisture content (>15% for wood, >1.0% for concrete). Common sources include plumbing leaks, roof leaks, condensation issues, poor drainage, and inadequate ventilation. All moisture problems must be permanently resolved before remediation can be successful.
- **Enhanced Cleaning Protocol:** Perform additional HEPA vacuuming and detailed cleaning of all surfaces within the containment area. This includes HEPA vacuuming of walls, ceilings, floors, and all horizontal surfaces. Follow with damp-wiping using appropriate antimicrobial solutions. Ensure all cleaning equipment is equipped with HEPA filtration to prevent cross-contamination.
- **Air Scrubbing and Negative Pressure:** Operate HEPA-filtered air scrubbers continuously during remediation activities to reduce airborne fungal particulates. Maintain negative air pressure within the containment area relative to adjacent spaces to prevent migration of contaminants. Air scrubbers should achieve a minimum of 4-6 air changes per hour and be operated for at least 24-48 hours after completion of cleaning activities.
- **Follow-Up Clearance Testing:** Conduct follow-up post-remediation clearance testing after all corrective measures have been completed and the area has been thoroughly cleaned. Testing should include both indoor and outdoor air samples using the same methodology as the initial assessment. Clearance testing should not be performed until all remediation work is complete, the area has been cleaned, and sufficient time has elapsed for airborne spore levels to stabilize (typically 24-48 hours after final cleaning).
- **Professional Remediation Services:** Engage qualified, licensed remediation professionals with experience in mold remediation and knowledge of industry standards (IICRC S520). Ensure all personnel are properly trained and use appropriate personal protective equipment. Remediation should follow established containment, removal, and cleaning protocols to prevent cross-contamination and ensure worker safety.

Note: All recommendations should be implemented under the supervision of qualified professionals and in accordance with applicable industry guidelines (IICRC S520, EPA Mold Remediation Guidelines, OSHA standards). The property should not be occupied during active remediation work, and clearance must be achieved before returning to normal use.

X. LABORATORY INFORMATION

Laboratory: Southeast Environmental Microbiology Laboratories

Accreditation: AIHA LAP, LLC — TX LAB1016

XI. LIMITATIONS

This report is subject to limitations inherent in mold inspections, including the potential for hidden mold growth and the inability to identify all conditions without invasive procedures. The findings are based on the conditions observed at the time of inspection and do not account for future changes.

OVERALL ASSESSMENT

The property exhibits evidence of fungal amplification and elevated spore concentrations. **Corrective action and remediation are recommended** to address affected materials and restore indoor air quality.

ELEVATED CONDITIONS FOUND / FAILED CLEARANCE



CERTIFICATION

License #: 14447000

Rapid Response Mold Testing

+92 03168614915

Moldtesting@gmail.com

www.company.com

LIMITATION OF LIABILITY & DISCLAIMER: This report is based on the conditions observed at the time of the inspection. **Rapid Response Mold Testing** does not guarantee or warrant that all mold or moisture issues have been identified. The findings are based on professional judgment, visible evidence, and sampling results. **Rapid Response Mold Testing** shall not be liable for any damages, including but not limited to direct, indirect, incidental, or consequential damages arising from the use of this report. This report is for the exclusive use of the client named herein and is not transferable. Remediation should always be performed by a licensed professional in accordance with IICRC S520 standards.

End of Report

Southeast Environmental Microbiology Laboratories

410 W Grand Pkwy S, Suite 250
Katy, TX. 77494
Phone: 832-437-2667

The information and data for **Rapid Response Mold Testing** has been checked for thoroughness and accuracy. The following reports are contained within this document:

☐ Surface/Bulk Report

☐ Andersen Fungal Report

☒ Spore Trap Report

☐ Quantitative Fungal Report

Lab Manager Review :

Magzoub Ismail

Date : 06-23-2026

Thank you for using SEEML laboratories. We strive to provide superior quality and service. SEEML laboratories are accredited through AIHA LAP, LLC (EMLAP #232339) for the analysis of Spore Traps and Surface/Bulk Samples and licensed by the Texas Department of Licensing and Regulation (LAB1016).

The data within this report is reliable to three significant figures. The third significant figure is technically unjustified. In this instance, the third figure is reported as an estimate to facilitate the interpretation by the customer.

Confidentiality Notice:

The document(s) contained herein are confidential and privileged information, intended for the exclusive use of the individual or entity named above. If the reader of this message is not the intended recipient, or the employee or agent responsible for delivering it to the intended recipient, you are hereby notified that any dissemination, distribution or copying of the document(s) is strictly prohibited. If you have received this document in error, please immediately notify us by telephone to arrange for its return. Thank you.

Guidelines for Interpretation:

No accepted quantitative regulatory standards currently exist by which to assess the health risks related to mold and bacterial exposure. Molds and bacteria have been associated with a variety of health effects and sensitivity varies from person to person.

Several organizations, including: the American Conference of Government Industrial Hygienists (ACGIH); the American Industrial Hygiene Association (AIHA); the Indoor Air Quality Association (IAQA); the United States Environmental Protection Agency (USEPA); the Centers for Disease Control (CDC), as well as the California Department of Health Services (CADHS), have all published guidelines for assessment and interpretation of mold resulting from water intrusion in buildings.

Interpretation of the data and information within this document is left to the company, consultant, and/or persons who conducted the fieldwork.

Spore Trap Report

Rapid Response Mold Testing				Date Sampled: 06/20/2026			
8204 Atlantic Ave				Date Received: 06/23/2026			
Cudahy, CA, 90201				Date Analyzed: 06/23/2026			
(310) 774-7584				Date Reported: 06/23/2026			
				Date Revised:			
				Project Name: Kala Restoration			
				Project Number:			
				Project Address: 645 Loranne Ave			
				Project City, State, ZIP: Pomona, CA 91767			
TEST METHOD: DIRECT MICROSCOPY EXAMINATION SEEML SOP 7				SEEML Reference # : H-260623102			
Client Sample ID	4247 9207			4247 9208			
Location	Master Bedroom			Backyard			
Lab Sample ID	H-260623102-434			H-260623102-435			
Detection Limit (spores/m ³)	13			13			
Hyphal Fragments							
Pollen							
Spore Trap Used	AOC			AOC			
	raw ct.	spores/m ³	%	raw ct.	spores/m ³	%	
Alternaria (=Ulocladium)							
Ascospores	16	213	8				
Basidiospores	20	267	9				
Bipolaris/Drechslera	1	13	<1				
Cercospora							
Chaetomium							
Cladosporium	148	1973	69	12	160	67	
Colorless/Other Brown*							
Curvularia							
Epicoccum	2	27	1	2	27	11	
Fusarium							
Memnoniella							
Nigrospora							
Oidium							
Penicillium/Aspergillus	20	267	9	4	53	22	
Pithomyces	1	13	<1				
Polythrincium							
Pyricularia							
Rusts							
Smuts/Periconia/Myxomy							
Spegazzinia							
Stachybotrys							
Tetraploa	2	27	1				
Torula	3	40	1				
Zygomycetes							
Background debris (1-5)**	3			3			
Sample Volume(liters)	75			75			
TOTAL SPORES/M³	213	2840		18	240		

Comments:

Spore types listed without a count or data entry were not detected during the course of the analysis for the respective sample, indicating a raw count of <1 spore.

The analytical sensitivity is the spores/m³ divided by the raw count, expressed in spores/m³. The limit of detection is the analytical sensitivity

(in spores/m³) multiplied by the sample volume (in liters) divided by 1000 liters.

*Colorless, other Brown are spores without a distinctive morphology on spore traps and non-viable surface samples.

**Background debris is the amount of particulate matter present on the slide and is graded from 1-5 with 1 = very light, 2= Light, 3 = Medium, 4 = Heavy, 5 = Very Heavy. The higher the rating the more likelihood spores may be underestimated. A rating of 5 should be interpreted as minimal counts and may actually be higher than reported.

***Ulocladium has been recognized by the International Mycological Association to be equal to Alternaria and so they are reported as one.

Disclaimer: The sample results are determined by the sample volume, which is provided by the customer.

This report relates only to the samples tested as they were received.

Respectfully submitted, SEEML

Magzoub Ismail

Magzoub Ismail, Approved Laboratory Signatory

410 W Grand Pkwy S, Suite 250

Katy, TX. 77494

Phone: 832-437-2667

Fungal Descriptions

Alternaria sp.

Aw - 0.89. Conidia dimensions: 18-83 x 7-18 microns. A very common allergen with an IgE mediated response. It is often found in carpets, textiles and on horizontal surfaces in building interiors. Often found on window frames. Outdoors it may be isolated from samples of soil, seeds and plants. It is commonly found in outdoor samples. The large spore size, 20 - 200 microns in length and 7 - 18 microns in sizes, suggests that the spores from these fungi will be deposited in the nose, mouth and upper respiratory tract. It may be related to bakers' asthma. It has been associated with hypersensitivity pneumonitis.

The species *Alternaria alternata* can produce tenuazonic acid and other toxic metabolites that may be associated with disease in humans or animals. Common cause of extrinsic asthma (immediate-type hypersensitivity: type I). Acute symptoms include edema and bronchospasms; chronic cases may develop pulmonary emphysema.

Ascospore

A spore borne in a special cell called an ascus. Spores of this type are reported to be allergenic. All ascomycetes, members of a group of fungi called Ascomycotina, have this type of spore. The minute black dots on rotting wood and leaves or the little cups on lichens are examples of ascomycetes; another is the "truffle" mushroom.

Aspergillus/Penicillium

These are two of the most commonly found allergenic fungi in problem buildings. *Aspergillus* comes in many varieties (species). Many of the varieties produce toxic substances. It may be associated with symptoms such as sinusitis, allergic bronchopulmonary aspergillosis, and other allergic symptoms. *Penicillium* is a variety of mold that is very common indoors and is found in increased numbers in problem buildings. It also has many varieties, some of which produce toxic substances. The symptoms are allergic reactions, mucous membrane irritation, headaches, vomiting, and diarrhea. Due to the morphological similarity of *Aspergillus* and *Penicillium*, they are not differentiated by microscopic analysis and are reported together.

Aspergillus sp.

Aw 0.75 - 0.82. Reported to be allergenic. Members of this genus are reported to cause ear infections. Many species produce mycotoxins that may be associated with disease in humans and other animals. Toxin production is dependent on the species or a strain within a species and on the food source for the fungus. Some of these toxins have been found to be carcinogenic in animal species. Several toxins are considered potential human carcinogens. Common cause of extrinsic asthma (immediate-type hypersensitivity: type I). Acute symptoms include edema and bronchospasms; chronic cases may develop pulmonary emphysema; may also be associated with sinusitis, allergic bronchopulmonary aspergillosis, and other allergic symptoms.

Aureobasidium

Aureobasidium pullulans is a ubiquitous and generalistic black, yeast-like fungus that can be found in different environments (e.g., soil, water, air, and limestone). It is well known as a naturally occurring epiphyte or endophyte of a wide range of plant species (e.g. apple, grape, cucumber, green beans, cabbage) without causing any symptoms of disease. *A. pullulans* has a high importance in biotechnology for the production of different enzymes, siderophores and pullulan. Furthermore, *A. pullulans* is used in biological control of plant diseases, especially storage diseases.

Chronic human exposure to *A. pullulans* via humidifiers or air conditioners can lead to hypersensitivity pneumonitis (extrinsic allergic alveolitis) or "humidifier lung". This condition is characterized acutely by dyspnea, cough, fever, chest infiltrates, and acute inflammatory reaction. The condition can also be chronic, and lymphocyte mediated. The chronic condition is characterized radiographically by reticulonodular infiltrates in the lung, with apical sparing. The strains causing infections in humans were reclassified to *A. melanogenum*.

Basidiospore

Spore from basidiomycetes. Many varieties are reported to be allergenic.

Bipolaris sp.

Aw 0.93. Conidia dimensions: 7-14 x 5-9 microns. It is parasitic on plants and soft fruits. Found in soil and on house plants and vegetables, it is also known as "gray mold". It causes leaf rot on grapes, strawberries, lettuce, etc. It is a well-known allergen, producing asthma type symptoms in greenhouse workers and "wine grower's lung".

Botrytis sp.

A fungus with large spores that could be expected to be deposited in the upper respiratory tract. This fungus can produce the mycotoxin - sterigmatocystin, which has been shown to produce liver and kidney damage when ingested by laboratory animals.

Cercospora

Common outdoors in agricultural areas, especially during harvest. Parasite of higher plants, causing leaf spot. Commonly found as parasites on higher plants.

Chaetomium sp.

Large ascomycetous fungus producing perithecia. It is found on a variety of substrates containing cellulose, including paper and plant compost. It has been found on paper in sheetrock. It can produce an *Acremonium*-like state on fungal media. Varieties are considered allergenic and have been associated with peritonitis, cutaneous lesions, and system mycosis.

Cladosporium sp.

Aw 0.88; Aw 0.84. Most commonly identified outdoor fungus. The outdoor numbers are reduced in the winter. The numbers are often high in the summer. Often found indoors in numbers less than outdoor numbers. It is a common allergen. Indoor *Cladosporium* sp. may be different than the species identified outdoors. It is commonly found on the surface of fiberglass duct liners in the interior of supply ducts. A wide variety of plants are food sources for this fungus. It is found on dead plants, woody plants, food, straw, soil, paint, and textiles. Produces greater than 10 antigens. Antigens in commercial extracts are of variable quality and may degrade within weeks of preparation. Common cause of extrinsic asthma (immediate-type hypersensitivity: type I). Acute symptoms include skin lesions, eye ulceration, mycosis (including onychomycosis, an infection of the nails of the feet or hands) edema and bronchospasms; chronic cases may develop pulmonary emphysema.

Curvularia sp.

Reported to be allergenic and has been associated with allergic fungal sinusitis. It may cause corneal infections, mycetoma, and infections in immune compromised hosts.

Drechslera sp.

Conidia dimensions: 40-120 x 17-28 microns. Found on grasses, grains and decaying food. It can occasionally cause a corneal infection of the eye.

Epicoccum sp.

Conidia dimensions: 15-25 microns. A common allergen. It is found in plants, soil, grains, textiles and paper products.

Fusarium sp.

Aw 0.90. A common soil fungus. It is found on a wide range of plants. It is often found in humidifiers. Several species in this genus can produce potent trichothecene toxins. The trichothecene (scirpene) toxin targets the following systems: circulatory, alimentary, skin, and nervous. Produces vomitoxin on grains during unusually damp growing conditions. Symptoms may occur either through ingestion of contaminated grains or possibly inhalation of spores. The genera can produce hemorrhagic syndrome in humans (alimentary toxic aleukia). This is characterized by nausea, vomiting, diarrhea, dermatitis, and extensive internal bleeding. Reported to be allergenic. Frequently involved in eye, skin, and nail infections.

Myxomycetes

Members of a group of fungi that is included in the category of "slime molds". They're occasionally found indoors, but mainly reside in forested regions on decaying logs, stumps, and dead leaves. Myxomycetes display characteristics of fungi *and* protozoans. In favorable (wet) conditions they exhibit motile, amoeba-like cells, usually bounded only by a plasma membrane, that are variable in size and form. During dry spells, they form a resting body (sclerotium) with dry, airborne spores. These fungi are not known to produce toxins but can cause hay fever and asthma.

Memnoniella

Contaminant found most often with *Stachybotrys* on wet cellulose. Forms in chains, but it are very similar to *Stachybotrys* and sometimes is considered to be in the *Stachybotrys* family. Certain species do produce toxins very similar to the ones produced by *Stachybotrys chartarum* and many consider the IAQ importance of *Memnoniella* to be on par with *Stachybotrys*. Allergenic and infectious properties are not well studied.

Nigrospora sp.

Commonly found in warm climates, this mold may be responsible for allergic reactions such as hay fever and asthma. It is found on decaying plant material and in the soil. It is not often found indoors.

Oidium sp.

The asexual phase of *Erysiphe* sp. It is a plant pathogen causing powdery mildews. It is very common on the leaf's stems, and flowers of plants. The health effects and allergenicity have not been studied. It does not grow on non-living surfaces such as wood or drywall.

Penicillium sp.

Aw 0.78 - 0.88. A wide number of organisms have been placed in this genus. Identification to species is difficult. Often found in aerosol samples. Commonly found in soil, food, cellulose and grains. It is also found in paint and compost piles. It may cause hypersensitivity pneumonitis, allergic alveolitis in susceptible individuals. It is reported to be allergenic (skin). It is commonly found in carpet, wallpaper, and in interior fiberglass duct insulation. Some species can produce mycotoxins. Common cause of extrinsic asthma (immediate-type hypersensitivity: type I). Acute symptoms include edema and bronchospasms; chronic cases may develop pulmonary emphysema. It may also cause headaches, vomiting, and diarrhea.

Periconia sp.

Periconia sp. are found in soil, blackened and dead herbaceous stems leaf spots, grasses, rushes, and sedges. Almost always associated with other fungi. Rarely found growing indoors. Reportedly associated with a rare case of mycotic keratitis.

Pithomyces sp.

A common mold found on dead leaves, plants, soil and especially grasses. Causes facial eczema in ruminants. It exhibits distinctive multi-celled brown conidia. It is not known to be a human allergen or pathogen. It is rarely found indoors, although it can grow on paper.

Rusts/Smuts

These fungi are associated with plant diseases. In the classification scheme of the fungi, the smuts have much in common with the rusts, and they are frequently discussed together. Both groups produce wind-borne, resistant teliospores that serve as the basis for their classification and their means of spread. Rusts usually attack vegetative regions (i.e., leaves and stems) of plants; smuts usually are associated with the reproductive structures (seeds). They can cause hay fever and asthma.

Spegazzinia

Spegazzinia species comprise a very small proportion of the fungal biota. This genus is somewhat related to other lobed or ornamented genera such as *Candelabrum*. No information is available regarding health effects or toxicity. Allergenicity has not been studied. Usually identified on spore trap samples where it is seen every few weeks. (Spores have very distinctive morphology.) May also be found in air by culturable (Andersen) samples if a long enough incubation period is provided so that sporulation occurs. Our laboratory has never found this organism growing on indoor environmental surfaces. Natural habitat includes soil and many kinds of trees and plants.

Sporotrichum

Sporotrichum species comprise a very small proportion of the fungal biota and are most closely related to other colorless hyphomycetes such as *Chrysosporium*. The only information available regarding health effects are a few rare cases of repeated isolations from respiratory secretions suggestive of bronchopulmonary colonization. No information is available regarding toxicity. Allergenicity has not been studied. May be identified on surfaces by tape lifts, tease mounts from bulk samples, and in air by culturable (Andersen) samples. Many times sporulating colorless fungi are very difficult to identify, with critical structures only very faintly visible under oil immersion magnification. Thus, these isolates may be placed in the category "Colorless, sporulating, ID unknown" on the Andersen report format. Spores do not have distinctive morphology and would be categorized as "other colorless" on spore trap samples. Natural habitat includes soil and decaying wood.

Stachybotrys sp.

Aw - 0.94, optimum Aw $\rightarrow$ 0.98. Several strains of this fungus (*S. atra*, *S. chartarum* and *S. alternans* are synonymous) may produce a trichothecene mycotoxin- Satratoxin H - which is poisonous by inhalation. The toxins are present on the fungal spores. This is a slow growing fungus on media. It does not compete well with other rapidly growing fungi. The dark colored fungus grows on building material with high cellulose content and low nitrogen content. Areas with a relative humidity above 55%, and are subject to temperature fluctuations, are ideal for toxin production. Individuals with chronic exposure to the toxin produced by this fungus reported cold and flu symptoms, sore throats, diarrhea, headaches, fatigue, dermatitis, intermittent local hair loss and generalized malaise. Other symptoms include coughs, rhinitis, nosebleed, a burning sensation in the nasal passages, throat, and lungs, and fever. The toxins produced by this fungus will suppress the immune system affecting the lymphoid tissue and the bone marrow. Animals injected with the toxin from this fungus exhibited the following symptoms: necrosis and hemorrhage within the brain, thymus, spleen, intestine, lung, heart, lymph node, liver, and kidney. Effects by absorption of the toxin in the human lung are known as pneumomycosis.

This organism is rarely found in outdoor samples. It is usually difficult to find in indoor air samples unless it is physically disturbed (or possibly -this is speculation- a drop in the relative humidity). The spores are in a gelatinous mass. Appropriate media for the growth of this organism will have high cellulose content and low nitrogen content. The spores will die readily after release. The dead spores are still allergenic and toxigenic. Percutaneous absorption has caused mild symptoms.

Stemphylium sp.

Reported to be allergenic. Isolated from dead plants and cellulose materials.

Torula sp.

Found outdoors in air, soil, on dead vegetation, wood, and grasses. Also found indoors on cellulose materials. Reported to be allergenic and may cause hay fever and asthma.

Tetraploa

Tetraploa species comprise a very small proportion of the fungal biota. This genus is somewhat related to *Triposporium* and *Diplocladiella*. The only reported human infections are two cases of keratitis (1970, 1980) and one case of subcutaneous infection of the knee (1990). No information is available regarding other health effects or toxicity. Allergenicity has not been studied. Usually identified on spore trap samples where it is seen every few weeks. (Spores have very distinctive morphology.) Our laboratory has never found this organism growing on indoor environmental surfaces. Natural habitat includes leaf bases and stems just above the soil on many kinds of plants and trees.

Ulocladium sp.

Aw 0.89. Isolated from dead plants and cellulose materials. Found on textiles.

Zygomycetes

Zygomycetes are one of the four major groups of fungi, the others being the Oomycetes, the Ascomycetes, and the Basidiomycetes. Zygomycetes are common, fast growing, and often overgrow and/or inhibit other fungi nearby. *Rhizopus* and *Mucor* are two of the most common Zygomycetes seen in the indoor environment. However, others are seen as well, including *Syncephalastrum*, *Circinella*, *Mortierella*, *Mycotypha*, *Cunninghamella*, and *Choanephora*. For further information, please see descriptions of these individual genera.

The following table lists mycotoxins that are produced by certain types of fungi:

Fungi	Mycotoxin
<i>Acremonium crocinigenum</i>	Crocin
<i>Aspergillus favus</i>	Alfatoxin B, cyclopiazonic acid
<i>Aspergillus fumigatus</i>	Fumagilin, gliotoxin
<i>Aspergillus carneus</i>	Citrinin
<i>Aspergillus clavatus</i>	Cytochalasin, patulin
<i>Aspergillus Parasiticus</i>	Alfatoxin B
<i>Aspergillus nomius</i>	Alfatoxin B
<i>Aspergillus niger</i>	Ochratoxin A, malformin, oxalic acid
<i>Acremonium crocinigenum</i>	Crocin
<i>Aspergillus nidulans</i>	Sterigmatocystin
<i>Aspergillus ochraceus</i>	Ochratoxin A, penicillic acid
<i>Aspergillus versicolor</i>	Sterigmatocystin, 5 ethoxysterigmatocystin
<i>Aspergillus ustus</i>	Ausdiol, austamide, austocystin, brevianamide
<i>Aspergillus terreus</i>	Citreoviridin
<i>Alternaria</i>	Alternariol, altertoxin, altenuene, altenusin, tenuazonic acid
<i>Arthrimum</i>	Nitropropionic acid
<i>Bioploaris</i>	Cytochalasin, sporidesmin, sterigmatocystin
<i>Chaetomium</i>	Chaetoglobosin A,B,C. Sterigmatocystin
<i>Cladosporium</i>	Cladosporic acid
<i>Clavipes purpurea</i>	Ergotism
<i>Cylindrocoryn</i>	Trichothecene
<i>Diplodia</i>	Diplodiatxin
<i>Fusarium</i>	Trichothecene, zearalenone
<i>Fusarium moniliforme</i>	Fumonisin
<i>Emericella nidulans</i>	Sterigmatocystin
<i>Gliocladium</i>	Gliotoxin
<i>Memnoniella</i>	Griseofulvin, dechlorogriseofulvin, epi-dechlorogriseofulvin, trichodermin, trichodermol
<i>Myrothecium</i>	Trichothecene
<i>Paecilomyces</i>	Patulin, viriditoxin
<i>Penicillium aurantiocandidum</i>	Penicillic acid
<i>Penicillium aurantiogriseum</i>	Penicillic acid
<i>Penicillium brasilanum</i>	Penicillic acid
<i>Penicillium brevicompactum</i>	Mycophenolic acid
<i>Penicillium camemberti</i>	Cyclopiazonic acid
<i>Penicillium carneum</i>	Mycophenolic acid, Roquefortine C
<i>Penicillium crateriforme</i>	Rubratxin

Fungi	Mycotoxin
<i>Penicillium citrinum</i>	Citrinin
<i>Penicillium commune</i>	Cyclopiazonic acid
<i>Penicillium crustosum</i>	Roquefortine C
<i>Penicillium chrysogenum</i>	Roquefortine C
<i>Penicillium discolor</i>	Chaetoglobosin C
<i>Penicillium expansum</i>	Citrinin, Roquefortine C
<i>Penicillium griseofulvum</i>	Roquefortine C, cyclopiazonic acid, griseofulvin
<i>Penicillium hirsutum</i>	Roquefortine C
<i>Penicillium hordei</i>	Roquefortine C
<i>Penicillium nordicum</i>	Ochratoxin A
<i>Penicillium paneum</i>	Roquefortine C
<i>Penicillium palitans</i>	Cyclopiazonic acid
<i>Penicillium polonicum</i>	Penicillic acid
<i>Penicillium roqueforti</i>	Roquefortine C, Mycophenolic acid
<i>Penicillium veridicatum</i>	Penicillic acid
<i>Penicillium verrucosum</i>	Citrinin, ochratoxin A
<i>Penicillium/ Aspergillus</i>	Patulin
<i>Penicillium/ Aspergillus/Alternaria</i>	Glitoxin
<i>Phomopsis</i>	Macrocyclic trichothecenes
<i>Phoma</i>	Brefeldin, cytochalasin, secalonin acid, tenuazonic acid
<i>Pithomyces</i>	Sporidesmin
<i>Rhizoctonia</i>	Slaframine
<i>Rhizopus</i>	Rhizonin
<i>Sclerotinia</i>	Furanocoumarins
<i>Stachybotrys chartarum</i>	Iso-satratoxin F, roridin E, L-2, satratoxin G & H, trichodermin, trichodermol, trichothecene
<i>Torula</i>	Cytotoxins
<i>Trichoderma</i>	Trichodermin, trichodermol, gliotoxin
<i>Trichothecium</i>	Trichothecene
<i>Wallemia</i>	Wallemiol
<i>Zygosporium</i>	Cytochalasin

General terms

Allergen

An allergen is a substance that elicits an IgE antibody response and is responsible for producing allergic reactions. Chemicals are released when IgE on certain cells contact an allergen. These chemicals can cause injury to surrounding tissue - the visible signs of an allergy. Only a few fungal allergens have been characterized but all fungi are thought to be potentially allergenic. Fungal allergens are proteins found in either the mycelium or spores

"Black mold"

A poorly defined term. Black mold or toxic black mold has usually been associated with the mold *Stachybotrys chartarum*. While there are only a few molds that are truly black, there are many that can appear black. Not all molds that appear to be black are *Stachybotrys*.

Fungi

Fungi are neither animals nor plants and are classified in a kingdom of their own. The Kingdom of Fungi. Fungi include a very large group of organisms, including molds, yeasts, mushrooms and puffballs. There are >100,000 accepted fungal species but current estimates range to 1.5 million species. Mycologists (people who study fungi) have grouped fungi into four large groups according to their method of reproduction.

Hidden mold

This refers to visible mold growth on building structures that is not easily seen, including the areas above drop ceilings, within a wall cavity (the space between the inner and outer structure of a wall), inside air handlers, or within the ducting of a heating/ventilation system.

Microbial Volatile Organic Compounds (MVOCs)

Fungi produce chemicals as a result of their metabolism. Some of these chemicals, MVOCs, are responsible for the characteristic moldy, musty, or earthy smell of fungi, whether mushrooms or molds. Some MVOCs are considered offensive or annoying. Specific MVOCs are thought to be characteristic of wood rot and mold growth on building materials. The human nose is very sensitive to mold odors and sometimes more so than current analytical instruments.

Mold

Molds are a group of organisms that belong to the Kingdom of Fungi (see Fungi). Even though the terms mold and fungi had been commonly referred to interchangeably, all molds are fungi, but not all fungi are molds.

Mycotoxin

Mycotoxins are compounds produced by some fungi that are toxic to humans or animals. By convention, the term? Mycotoxin. Excludes mushroom toxins. Fungi that produce mycotoxins are called "toxigenic fungi."

Spore

General term for a reproductive structure in fungi, bacteria and some plants. In fungi, the spore is the structure which may be used for dissemination and may be resistant to adverse environmental conditions.

Toxic mold

The term "toxic mold" has no scientific meaning since the mold itself is not toxic. The metabolic byproducts of some molds may be toxic (see mycotoxin).

Hypha (plural, hyphae)

An individual fungal thread or filament of connected cells; the thread that represents the individual parts of the fungal body.

Southeast Environmental Microbiology Laboratories

410 W Grand Pkwy S, Suite 250
Katy, TX. 77494
Phone: 832-437-2667

The information and data for **Rapid Response Mold Testing** has been checked for thoroughness and accuracy. The following reports are contained within this document:

☐ Surface/Bulk Report

☐ Andersen Fungal Report

☒ Spore Trap Report

☐ Quantitative Fungal Report

Lab Manager Review : *Magzoub Ismail* Date : 06-23-2026

Thank you for using SEEML laboratories. We strive to provide superior quality and service. SEEML laboratories are accredited through AIHA LAP, LLC (EMLAP #232339) for the analysis of Spore Traps and Surface/Bulk Samples and licensed by the Texas Department of Licensing and Regulation (LAB1016).

The data within this report is reliable to three significant figures. The third significant figure is technically unjustified. In this instance, the third figure is reported as an estimate to facilitate the interpretation by the customer.

Confidentiality Notice:

The document(s) contained herein are confidential and privileged information, intended for the exclusive use of the individual or entity named above. If the reader of this message is not the intended recipient, or the employee or agent responsible for delivering it to the intended recipient, you are hereby notified that any dissemination, distribution or copying of the document(s) is strictly prohibited. If you have received this document in error, please immediately notify us by telephone to arrange for its return. Thank you.

Guidelines for Interpretation:

No accepted quantitative regulatory standards currently exist by which to assess the health risks related to mold and bacterial exposure. Molds and bacteria have been associated with a variety of health effects and sensitivity varies from person to person.

Several organizations, including: the American Conference of Government Industrial Hygienists (ACGIH); the American Industrial Hygiene Association (AIHA); the Indoor Air Quality Association (IAQA); the United States Environmental Protection Agency (USEPA); the Centers for Disease Control (CDC), as well as the California Department of Health Services (CADHS), have all published guidelines for assessment and interpretation of mold resulting from water intrusion in buildings.

Interpretation of the data and information within this document is left to the company, consultant, and/or persons who conducted the fieldwork.

Spore Trap Report

Rapid Response Mold Testing				Date Sampled: 06/20/2026			
8204 Atlantic Ave				Date Received: 06/23/2026			
Cudahy, CA, 90201				Date Analyzed: 06/23/2026			
(310) 774-7584				Date Reported: 06/23/2026			
				Date Revised:			
				Project Name: Kala Restoration			
				Project Number:			
				Project Address: 645 Loranne Ave			
				Project City, State, ZIP: Pomona, CA 91767			
TEST METHOD: DIRECT MICROSCOPY EXAMINATION SEEML SOP 7				SEEML Reference #: H-260623102			
Client Sample ID	4247 9207			4247 9208			
Location	Master Bedroom			Backyard			
Lab Sample ID	H-260623102-434			H-260623102-435			
Detection Limit (spores/m ³)	13			13			
Hyphal Fragments							
Pollen							
Spore Trap Used	AOC			AOC			
	raw ct.	spores/m ³	%	raw ct.	spores/m ³	%	
Alternaria (=Ulocladium)							
Ascospores	16	213	8				
Basidiospores	20	267	9				
Bipolaris/Drechslera	1	13	<1				
Cercospora							
Chaetomium							
Cladosporium	148	1973	69	12	160	67	
Colorless/Other Brown*							
Curvularia							
Epicoccum	2	27	1	2	27	11	
Fusarium							
Memnoniella							
Nigrospora							
Oidium							
Penicillium/Aspergillus	20	267	9	4	53	22	
Pithomyces	1	13	<1				
Polythrincium							
Pyricularia							
Rusts							
Smuts/Periconia/Myxomy							
Spegazzinia							
Stachybotrys							
Tetraploa	2	27	1				
Torula	3	40	1				
Zygomycetes							
Background debris (1-5)**	3			3			
Sample Volume(liters)	75			75			
TOTAL SPORES/M³	213	2840		18	240		

Comments:

Spore types listed without a count or data entry were not detected during the course of the analysis for the respective sample, indicating a raw count of <1 spore.

The analytical sensitivity is the spores/m³ divided by the raw count, expressed in spores/m³. The limit of detection is the analytical sensitivity

(in spores/m³) multiplied by the sample volume (in liters) divided by 1000 liters.

*Colorless, other Brown are spores without a distinctive morphology on spore traps and non-viable surface samples.

**Background debris is the amount of particulate matter present on the slide and is graded from 1-5 with 1 = very light, 2= Light, 3 = Medium, 4 = Heavy, 5 = Very Heavy. The higher the rating the more likelihood spores may be underestimated. A rating of 5 should be interpreted as minimal counts and may actually be higher than reported.

***Ulocladium has been recognized by the International Mycological Association to be equal to Alternaria and so they are reported as one.

Disclaimer: The sample results are determined by the sample volume, which is provided by the customer.

This report relates only to the samples tested as they were received.

Respectfully submitted, SEEML

Magzoub Ismail

Magzoub Ismail, Approved Laboratory Signatory

410 W Grand Pkwy S, Suite 250

Katy, TX. 77494

Phone: 832-437-2667

Fungal Descriptions

Alternaria sp.

Aw - 0.89. Conidia dimensions: 18-83 x 7-18 microns. A very common allergen with an IgE mediated response. It is often found in carpets, textiles and on horizontal surfaces in building interiors. Often found on window frames. Outdoors it may be isolated from samples of soil, seeds and plants. It is commonly found in outdoor samples. The large spore size, 20 - 200 microns in length and 7 - 18 microns in sizes, suggests that the spores from these fungi will be deposited in the nose, mouth and upper respiratory tract. It may be related to bakers' asthma. It has been associated with hypersensitivity pneumonitis.

The species *Alternaria alternata* can produce tenuazonic acid and other toxic metabolites that may be associated with disease in humans or animals. Common cause of extrinsic asthma (immediate-type hypersensitivity: type I). Acute symptoms include edema and bronchospasms; chronic cases may develop pulmonary emphysema.

Ascospore

A spore borne in a special cell called an ascus. Spores of this type are reported to be allergenic. All ascomycetes, members of a group of fungi called Ascomycotina, have this type of spore. The minute black dots on rotting wood and leaves or the little cups on lichens are examples of ascomycetes; another is the "truffle" mushroom.

Aspergillus/Penicillium

These are two of the most commonly found allergenic fungi in problem buildings. *Aspergillus* comes in many varieties (species). Many of the varieties produce toxic substances. It may be associated with symptoms such as sinusitis, allergic bronchopulmonary aspergillosis, and other allergic symptoms. *Penicillium* is a variety of mold that is very common indoors and is found in increased numbers in problem buildings. It also has many varieties, some of which produce toxic substances. The symptoms are allergic reactions, mucous membrane irritation, headaches, vomiting, and diarrhea. Due to the morphological similarity of *Aspergillus* and *Penicillium*, they are not differentiated by microscopic analysis and are reported together.

Aspergillus sp.

Aw 0.75 - 0.82. Reported to be allergenic. Members of this genus are reported to cause ear infections. Many species produce mycotoxins that may be associated with disease in humans and other animals. Toxin production is dependent on the species or a strain within a species and on the food source for the fungus. Some of these toxins have been found to be carcinogenic in animal species. Several toxins are considered potential human carcinogens. Common cause of extrinsic asthma (immediate-type hypersensitivity: type I). Acute symptoms include edema and bronchospasms; chronic cases may develop pulmonary emphysema; may also be associated with sinusitis, allergic bronchopulmonary aspergillosis, and other allergic symptoms.

Aureobasidium

Aureobasidium pullulans is a ubiquitous and generalistic black, yeast-like fungus that can be found in different environments (e.g., soil, water, air, and limestone). It is well known as a naturally occurring epiphyte or endophyte of a wide range of plant species (e.g. apple, grape, cucumber, green beans, cabbage) without causing any symptoms of disease. *A. pullulans* has a high importance in biotechnology for the production of different enzymes, siderophores and pullulan. Furthermore, *A. pullulans* is used in biological control of plant diseases, especially storage diseases.

Chronic human exposure to *A. pullulans* via humidifiers or air conditioners can lead to hypersensitivity pneumonitis (extrinsic allergic alveolitis) or "humidifier lung". This condition is characterized acutely by dyspnea, cough, fever, chest infiltrates, and acute inflammatory reaction. The condition can also be chronic, and lymphocyte mediated. The chronic condition is characterized radiographically by reticulonodular infiltrates in the lung, with apical sparing. The strains causing infections in humans were reclassified to *A. melanogenum*.

Basidiospore

Spore from basidiomycetes. Many varieties are reported to be allergenic.

Bipolaris sp.

Aw 0.93. Conidia dimensions: 7-14 x 5-9 microns. It is parasitic on plants and soft fruits. Found in soil and on house plants and vegetables, it is also known as "gray mold". It causes leaf rot on grapes, strawberries, lettuce, etc. It is a well-known allergen, producing asthma type symptoms in greenhouse workers and "wine grower's lung".

Botrytis sp.

A fungus with large spores that could be expected to be deposited in the upper respiratory tract. This fungus can produce the mycotoxin - sterigmatocystin, which has been shown to produce liver and kidney damage when ingested by laboratory animals.

Cercospora

Common outdoors in agricultural areas, especially during harvest. Parasite of higher plants, causing leaf spot. Commonly found as parasites on higher plants.

Chaetomium sp.

Large ascomycetous fungus producing perithecia. It is found on a variety of substrates containing cellulose, including paper and plant compost. It has been found on paper in sheetrock. It can produce an *Acremonium*-like state on fungal media. Varieties are considered allergenic and have been associated with peritonitis, cutaneous lesions, and system mycosis.

Cladosporium sp.

Aw 0.88; Aw 0.84. Most commonly identified outdoor fungus. The outdoor numbers are reduced in the winter. The numbers are often high in the summer. Often found indoors in numbers less than outdoor numbers. It is a common allergen. Indoor *Cladosporium* sp. may be different than the species identified outdoors. It is commonly found on the surface of fiberglass duct liners in the interior of supply ducts. A wide variety of plants are food sources for this fungus. It is found on dead plants, woody plants, food, straw, soil, paint, and textiles. Produces greater than 10 antigens. Antigens in commercial extracts are of variable quality and may degrade within weeks of preparation. Common cause of extrinsic asthma (immediate-type hypersensitivity: type I). Acute symptoms include skin lesions, eye ulceration, mycosis (including onychomycosis, an infection of the nails of the feet or hands) edema and bronchospasms; chronic cases may develop pulmonary emphysema.

Curvularia sp.

Reported to be allergenic and has been associated with allergic fungal sinusitis. It may cause corneal infections, mycetoma, and infections in immune compromised hosts.

Drechslera sp.

Conidia dimensions: 40-120 x 17-28 microns. Found on grasses, grains and decaying food. It can occasionally cause a corneal infection of the eye.

Epicoccum sp.

Conidia dimensions: 15-25 microns. A common allergen. It is found in plants, soil, grains, textiles and paper products.

Fusarium sp.

Aw 0.90. A common soil fungus. It is found on a wide range of plants. It is often found in humidifiers. Several species in this genus can produce potent trichothecene toxins. The trichothecene (scirpene) toxin targets the following systems: circulatory, alimentary, skin, and nervous. Produces vomitoxin on grains during unusually damp growing conditions. Symptoms may occur either through ingestion of contaminated grains or possibly inhalation of spores. The genera can produce hemorrhagic syndrome in humans (alimentary toxic aleukia). This is characterized by nausea, vomiting, diarrhea, dermatitis, and extensive internal bleeding. Reported to be allergenic. Frequently involved in eye, skin, and nail infections.

Myxomycetes

Members of a group of fungi that is included in the category of "slime molds". They're occasionally found indoors, but mainly reside in forested regions on decaying logs, stumps, and dead leaves. Myxomycetes display characteristics of fungi *and* protozoans. In favorable (wet) conditions they exhibit motile, amoeba-like cells, usually bounded only by a plasma membrane, that are variable in size and form. During dry spells, they form a resting body (sclerotium) with dry, airborne spores. These fungi are not known to produce toxins but can cause hay fever and asthma.

Memnoniella

Contaminant found most often with *Stachybotrys* on wet cellulose. Forms in chains, but it are very similar to *Stachybotrys* and sometimes is considered to be in the *Stachybotrys* family. Certain species do produce toxins very similar to the ones produced by *Stachybotrys chartarum* and many consider the IAQ importance of *Memnoniella* to be on par with *Stachybotrys*. Allergenic and infectious properties are not well studied.

Nigrospora sp.

Commonly found in warm climates, this mold may be responsible for allergic reactions such as hay fever and asthma. It is found on decaying plant material and in the soil. It is not often found indoors.

Oidium sp.

The asexual phase of *Erysiphe* sp. It is a plant pathogen causing powdery mildews. It is very common on the leaf's stems, and flowers of plants. The health effects and allergenicity have not been studied. It does not grow on non-living surfaces such as wood or drywall.

Penicillium sp.

Aw 0.78 - 0.88. A wide number of organisms have been placed in this genus. Identification to species is difficult. Often found in aerosol samples. Commonly found in soil, food, cellulose and grains. It is also found in paint and compost piles. It may cause hypersensitivity pneumonitis, allergic alveolitis in susceptible individuals. It is reported to be allergenic (skin). It is commonly found in carpet, wallpaper, and in interior fiberglass duct insulation. Some species can produce mycotoxins. Common cause of extrinsic asthma (immediate-type hypersensitivity: type I). Acute symptoms include edema and bronchospasms; chronic cases may develop pulmonary emphysema. It may also cause headaches, vomiting, and diarrhea.

Periconia sp.

Periconia sp. are found in soil, blackened and dead herbaceous stems leaf spots, grasses, rushes, and sedges. Almost always associated with other fungi. Rarely found growing indoors. Reportedly associated with a rare case of mycotic keratitis.

Pithomyces sp.

A common mold found on dead leaves, plants, soil and especially grasses. Causes facial eczema in ruminants. It exhibits distinctive multi-celled brown conidia. It is not known to be a human allergen or pathogen. It is rarely found indoors, although it can grow on paper.

Rusts/Smuts

These fungi are associated with plant diseases. In the classification scheme of the fungi, the smuts have much in common with the rusts, and they are frequently discussed together. Both groups produce wind-borne, resistant teliospores that serve as the basis for their classification and their means of spread. Rusts usually attack vegetative regions (i.e., leaves and stems) of plants; smuts usually are associated with the reproductive structures (seeds). They can cause hay fever and asthma.

Spegazzinia

Spegazzinia species comprise a very small proportion of the fungal biota. This genus is somewhat related to other lobed or ornamented genera such as *Candelabrum*. No information is available regarding health effects or toxicity. Allergenicity has not been studied. Usually identified on spore trap samples where it is seen every few weeks. (Spores have very distinctive morphology.) May also be found in air by culturable (Andersen) samples if a long enough incubation period is provided so that sporulation occurs. Our laboratory has never found this organism growing on indoor environmental surfaces. Natural habitat includes soil and many kinds of trees and plants.

Sporotrichum

Sporotrichum species comprise a very small proportion of the fungal biota and are most closely related to other colorless hyphomycetes such as *Chrysosporium*. The only information available regarding health effects are a few rare cases of repeated isolations from respiratory secretions suggestive of bronchopulmonary colonization. No information is available regarding toxicity. Allergenicity has not been studied. May be identified on surfaces by tape lifts, tease mounts from bulk samples, and in air by culturable (Andersen) samples. Many times sporulating colorless fungi are very difficult to identify, with critical structures only very faintly visible under oil immersion magnification. Thus, these isolates may be placed in the category "Colorless, sporulating, ID unknown" on the Andersen report format. Spores do not have distinctive morphology and would be categorized as "other colorless" on spore trap samples. Natural habitat includes soil and decaying wood.

Stachybotrys sp.

Aw - 0.94, optimum Aw $\rightarrow$ 0.98. Several strains of this fungus (*S. atra*, *S. chartarum* and *S. alternans* are synonymous) may produce a trichothecene mycotoxin- Satratoxin H - which is poisonous by inhalation. The toxins are present on the fungal spores. This is a slow growing fungus on media. It does not compete well with other rapidly growing fungi. The dark colored fungus grows on building material with high cellulose content and low nitrogen content. Areas with a relative humidity above 55%, and are subject to temperature fluctuations, are ideal for toxin production. Individuals with chronic exposure to the toxin produced by this fungus reported cold and flu symptoms, sore throats, diarrhea, headaches, fatigue, dermatitis, intermittent local hair loss and generalized malaise. Other symptoms include coughs, rhinitis, nosebleed, a burning sensation in the nasal passages, throat, and lungs, and fever. The toxins produced by this fungus will suppress the immune system affecting the lymphoid tissue and the bone marrow. Animals injected with the toxin from this fungus exhibited the following symptoms: necrosis and hemorrhage within the brain, thymus, spleen, intestine, lung, heart, lymph node, liver, and kidney. Effects by absorption of the toxin in the human lung are known as pneumomycosis.

This organism is rarely found in outdoor samples. It is usually difficult to find in indoor air samples unless it is physically disturbed (or possibly -this is speculation- a drop in the relative humidity). The spores are in a gelatinous mass. Appropriate media for the growth of this organism will have high cellulose content and low nitrogen content. The spores will die readily after release. The dead spores are still allergenic and toxigenic. Percutaneous absorption has caused mild symptoms.

Stemphylium sp.

Reported to be allergenic. Isolated from dead plants and cellulose materials.

Torula sp.

Found outdoors in air, soil, on dead vegetation, wood, and grasses. Also found indoors on cellulose materials. Reported to be allergenic and may cause hay fever and asthma.

Tetraploa

Tetraploa species comprise a very small proportion of the fungal biota. This genus is somewhat related to *Triposporium* and *Diplocladiella*. The only reported human infections are two cases of keratitis (1970, 1980) and one case of subcutaneous infection of the knee (1990). No information is available regarding other health effects or toxicity. Allergenicity has not been studied. Usually identified on spore trap samples where it is seen every few weeks. (Spores have very distinctive morphology.) Our laboratory has never found this organism growing on indoor environmental surfaces. Natural habitat includes leaf bases and stems just above the soil on many kinds of plants and trees.

Ulocladium sp.

Aw 0.89. Isolated from dead plants and cellulose materials. Found on textiles.

Zygomycetes

Zygomycetes are one of the four major groups of fungi, the others being the Oomycetes, the Ascomycetes, and the Basidiomycetes. Zygomycetes are common, fast growing, and often overgrow and/or inhibit other fungi nearby. *Rhizopus* and *Mucor* are two of the most common Zygomycetes seen in the indoor environment. However, others are seen as well, including *Syncephalastrum*, *Circinella*, *Mortierella*, *Mycotypha*, *Cunninghamella*, and *Choanephora*. For further information, please see descriptions of these individual genera.

The following table lists mycotoxins that are produced by certain types of fungi:

Fungi	Mycotoxin
<i>Acremonium crocinigenum</i>	Crocin
<i>Aspergillus favus</i>	Alfatoxin B, cyclopiazonic acid
<i>Aspergillus fumigatus</i>	Fumagilin, gliotoxin
<i>Aspergillus carneus</i>	Citrinin
<i>Aspergillus clavatus</i>	Cytochalasin, patulin
<i>Aspergillus Parasiticus</i>	Alfatoxin B
<i>Aspergillus nomius</i>	Alfatoxin B
<i>Aspergillus niger</i>	Ochratoxin A, malformin, oxalic acid
<i>Acremonium crocinigenum</i>	Crocin
<i>Aspergillus nidulans</i>	Sterigmatocystin
<i>Aspergillus ochraceus</i>	Ochratoxin A, penicillic acid
<i>Aspergillus versicolor</i>	Sterigmatocystin, 5 ethoxysterigmatocystin
<i>Aspergillus ustus</i>	Ausdiol, austamide, austocystin, brevianamide
<i>Aspergillus terreus</i>	Citreoviridin
<i>Alternaria</i>	Alternariol, altertoxin, altenuene, altenusin, tenuazonic acid
<i>Arthrimum</i>	Nitropropionic acid
<i>Bioploaris</i>	Cytochalasin, sporidesmin, sterigmatocystin
<i>Chaetomium</i>	Chaetoglobosin A,B,C. Sterigmatocystin
<i>Cladosporium</i>	Cladosporic acid
<i>Clavipes purpurea</i>	Ergotism
<i>Cylindrocoryn</i>	Trichothecene
<i>Diplodia</i>	Diplodiatxin
<i>Fusarium</i>	Trichothecene, zearalenone
<i>Fusarium moniliforme</i>	Fumonisin
<i>Emmericella nidulans</i>	Sterigmatocystin
<i>Gliocladium</i>	Gliotoxin
<i>Memnoniella</i>	Griseofulvin, dechlorogriseofulvin, epi-dechlorogriseofulvin, trichodermin, trichodermol
<i>Myrothecium</i>	Trichothecene
<i>Paecilomyces</i>	Patulin, viriditoxin
<i>Penicillium aurantiocandidum</i>	Penicillic acid
<i>Penicillium aurantiogriseum</i>	Penicillic acid
<i>Penicillium brasilanum</i>	Penicillic acid
<i>Penicillium brevicompactum</i>	Mycophenolic acid
<i>Penicillium camemberti</i>	Cyclopiazonic acid
<i>Penicillium carneum</i>	Mycophenolic acid, Roquefortine C
<i>Penicillium crateriforme</i>	Rubratxin

Fungi	Mycotoxin
<i>Penicillium citrinum</i>	Citrinin
<i>Penicillium commune</i>	Cyclopiazonic acid
<i>Penicillium crustosum</i>	Roquefortine C
<i>Penicillium chrysogenum</i>	Roquefortine C
<i>Penicillium discolor</i>	Chaetoglobosin C
<i>Penicillium expansum</i>	Citrinin, Roquefortine C
<i>Penicillium griseofulvum</i>	Roquefortine C, cyclopiazonic acid, griseofulvin
<i>Penicillium hirsutum</i>	Roquefortine C
<i>Penicillium hordei</i>	Roquefortine C
<i>Penicillium nordicum</i>	Ochratoxin A
<i>Penicillium paneum</i>	Roquefortine C
<i>Penicillium palitans</i>	Cyclopiazonic acid
<i>Penicillium polonicum</i>	Penicillic acid
<i>Penicillium roqueforti</i>	Roquefortine C, Mycophenolic acid
<i>Penicillium veridicatum</i>	Penicillic acid
<i>Penicillium verrucosum</i>	Citrinin, ochratoxin A
<i>Penicillium/ Aspergillus</i>	Patulin
<i>Penicillium/ Aspergillus/Alternaria</i>	Glitoxin
<i>Phomopsis</i>	Macrocyclic trichothecenes
<i>Phoma</i>	Brefeldin, cytochalasin, secalonic acid, tenuazonic acid
<i>Pithomyces</i>	Sporidesmin
<i>Rhizoctonia</i>	Slaframine
<i>Rhizopus</i>	Rhizonin
<i>Sclerotinia</i>	Furanocoumarins
<i>Stachybotrys chartarum</i>	Iso-satratoxin F, roridin E, L-2, satratoxin G & H, trichodermin, trichodermol, trichothecene
<i>Torula</i>	Cytotoxins
<i>Trichoderma</i>	Trichodermin, trichodermol, gliotoxin
<i>Trichothecium</i>	Trichothecene
<i>Wallemia</i>	Walleminol
<i>Zygosporium</i>	Cytochalasin

General terms

Allergen

An allergen is a substance that elicits an IgE antibody response and is responsible for producing allergic reactions. Chemicals are released when IgE on certain cells contact an allergen. These chemicals can cause injury to surrounding tissue - the visible signs of an allergy. Only a few fungal allergens have been characterized but all fungi are thought to be potentially allergenic. Fungal allergens are proteins found in either the mycelium or spores

"Black mold"

A poorly defined term. Black mold or toxic black mold has usually been associated with the mold *Stachybotrys chartarum*. While there are only a few molds that are truly black, there are many that can appear black. Not all molds that appear to be black are *Stachybotrys*.

Fungi

Fungi are neither animals nor plants and are classified in a kingdom of their own. The Kingdom of Fungi. Fungi include a very large group of organisms, including molds, yeasts, mushrooms and puffballs. There are >100,000 accepted fungal species but current estimates range to 1.5 million species. Mycologists (people who study fungi) have grouped fungi into four large groups according to their method of reproduction.

Hidden mold

This refers to visible mold growth on building structures that is not easily seen, including the areas above drop ceilings, within a wall cavity (the space between the inner and outer structure of a wall), inside air handlers, or within the ducting of a heating/ventilation system.

Microbial Volatile Organic Compounds (MVOCs)

Fungi produce chemicals as a result of their metabolism. Some of these chemicals, MVOCs, are responsible for the characteristic moldy, musty, or earthy smell of fungi, whether mushrooms or molds. Some MVOCs are considered offensive or annoying. Specific MVOCs are thought to be characteristic of wood rot and mold growth on building materials. The human nose is very sensitive to mold odors and sometimes more so than current analytical instruments.

Mold

Molds are a group of organisms that belong to the Kingdom of Fungi (see Fungi). Even though the terms mold and fungi had been commonly referred to interchangeably, all molds are fungi, but not all fungi are molds.

Mycotoxin

Mycotoxins are compounds produced by some fungi that are toxic to humans or animals. By convention, the term? Mycotoxin. Excludes mushroom toxins. Fungi that produce mycotoxins are called "toxigenic fungi."

Spore

General term for a reproductive structure in fungi, bacteria and some plants. In fungi, the spore is the structure which may be used for dissemination and may be resistant to adverse environmental conditions.

Toxic mold

The term "toxic mold" has no scientific meaning since the mold itself is not toxic. The metabolic byproducts of some molds may be toxic (see mycotoxin).

Hypha (plural, hyphae)

An individual fungal thread or filament of connected cells; the thread that represents the individual parts of the fungal body.